

Matthew Snelgrove



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matthewSnelgrove
portfolio

Education

University of Toronto, Scarborough | Computer Science | CGPA: 3.72 September 2021 – Present
Courses: Data Structures, Algorithms, Software Design, Security, Databases, Operating Systems, Web Development

Experience

SS&C Technologies (IFDS) | Software Developer September 2023 – April 2024

- Eliminated 3 hours of managerial work per week by automating component dependency tracking with internal APIs, and visualized the output in graphical and tabular views in a modern React web app.
- Ensured the maintainability and extendibility of the project with modular design, jsDoc code documentation, and comprehensive wiki entries.
- Reduced defect response times by 50% by rapidly identifying and patching production bugs across team apps.
- Accelerated modernization projects by investigating internal tools and delivering actionable user guides to streamline their integration into existing systems.

Ontario Ministry of Transportation | Junior Technical Analyst January 2023 – April 2023

- Developed RESTful API for modernization of critical driver/vehicle registration systems with Java Spring Boot.
- Cut debugging time by over 50% through automated unit and integration testing with JUnit and Postman, achieving 100% code coverage and preemptively identifying errors caused by pending dependency updates.
- Recovered ~3 hours per week of developer time by building and deploying new versions with Maven and Azure Pipelines instead of time-consuming and error-prone manual builds.

Projects

Superbiddo   October 2024 – December 2024

- Implemented core features for the auction-based marketplace app by developing a flexible and scalable RESTful API using Express and a relational database using PostgreSQL.
- Optimized a Google Cloud VM to host the containerized app, achieving 100% uptime, cost efficiency, and secure networking through resource fine-tuning, proactive monitoring and alerts, and stringent network configurations.
- Eliminated 5 hours of deployment time weekly by automating consistent and reliable deployments with a bash script to build images, upload them to the VM, and start containers with secrets using a Docker Compose file.
- Mitigated OWSAP Top 10 web vulnerabilities by implementing session-based authentication and access control, salted password hashing, data validation and sanitization, CSRF protection, and HTTPS.

Pintos Operating System May 2024 – August 2024

- Extended skeleton code for a basic operating system in C to support system calls, indexed and extensible files, and ELF program execution while ensuring stability and performance in a multithreaded environment.
- Improved fairness and responsiveness by implementing priority donation and a multi-level feedback queue scheduler (MLFQS) to dynamically adjust thread priorities based on CPU usage and workload.
- Optimized memory management and reduced page fault frequency by 30% by designing a virtual memory system with LRU-approximated page replacement using the clock algorithm.

Assembly Racing Game  August 2022

- Developed a 2D racing game in MIPS Assembly with keyboard input and bitmap display output.
- Achieved 60 FPS gameplay by optimizing the game loop, sprite rendering, and collision detection algorithms.

Technical Skills

Languages: Java, JavaScript/TypeScript, Python, C/C++, C#, SQL (Postgres), Assembly
Frameworks & Libraries: Node.js, Express.js, Spring Boot, React, MongoDB, JUnit, jQuery
Developer Tools: Git, Docker, Google Cloud Platform, Azure, Postman, OpenApi